Coulomb Screening at an Energy Offset: Optimized Local Rates and Whole-Space Certificates

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Abstract

We analyze the local screening and observation interfaces in the manuscript *Uniform excess charge for Coulomb molecules and the outer radius of neutral atoms*. Keeping its actual conditional core, retained particles, Thomas–Fermi minimizer, and all-sector energy offset, we optimize its displayed patch budget. The normalized discrepancy gains become 1/11 in a possibly nuclear patch and 1/5 in a nucleus-free patch, instead of the source's convenient 0.02. These gains are optimal for the stated monomial budgets, not for the underlying quantum problem. Under explicit master-kernel and change-of-law hypotheses they give averaged-field transfer gains 1/77 and 1/35. We also derive a whole-space primal–dual certificate for the same local Thomas–Fermi gap, an $L^{5/3}$ smearing remainder for signed sources, and exact compact-source diagnostics of unmeasured energy. Classical Coulomb duality, Fenchel duality, Fourier analysis, and the hypercircle identity are credited rather than renamed. The source's molecular ionization and neutral-atom radius theorems are not independently established here. Contribution assessment: C3/5, source-specific optimized screening and conditional field-transfer rates, supported by whole-space error certificates while retaining the global quantum and multiscale hypotheses.

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1 The physical problem and the proof boundary

Let $M \geq 1$ distinct fixed nuclei have real charges $z_j \geq 1$ at $R_j \in \mathbb{R}^3$. Write $Z = \sum_j z_j$, $\nu = \sum_j z_j \delta_{R_j}$, $K(x) = |x|^{-1}$ and $V = K * \nu$. On the antisymmetric space $\bigwedge^n L^2(\mathbb{R}^3; \mathbb{C}^2)$ the source uses the quadratic form

$$H_n = \sum_{i=1}^{n} \left(-\tfrac{1}{2}\Delta_i - V(x_i) \right) + \sum_{i<j} |x_i - x_j|^{-1}, \qquad E_n = \inf \sigma(H_n), \quad E_0 = 0. \tag{1}$$

Its form domain is antisymmetric H^1 with two spin states. Nuclear repulsion is an omitted scalar independent of n. Strict binding means $E_n < E_{n-1}$, not merely existence of an eigenstate at a threshold. The source claims one universal C with strict binding only for $n \leq Z + CM$, uniformly over arbitrary distinct nuclear positions and charges. It also claims universal positive upper and lower bounds on $I_1(Z) = E(Z-1, Z) - E(Z, Z)$ and on

$$R(\Psi_Z) = \inf \left\{ r \geq 0 : \int_{|x|>r} \rho_{\Psi_Z}(x)\, \mathrm{d}x \leq \tfrac{1}{2} \right\} \tag{2}$$

for every normalized neutral ground state and integer $Z \geq 1$. Here $\int \rho_{\Psi_Z} = Z$. The radius retains an expected half-electron, not a fixed fraction of the density; no spherical symmetry or uniqueness of the ground state is assumed.

The source's coarse reduction gives a minimum $E = \min_{n \geq 0} E_n$ in a finite sector, using a reciprocal-potential number bound and sector monotonicity. A normalized finite-sector form vector ψ is a δ-state if $q_n[\psi] \leq E + \delta$. A ground state in its own sector need not have $\delta = 0$. This distinction is indispensable in event conditioning and in the return to neutral sectors.

1.1 What is retained and what is added

The following dependency map separates the local work below from the long source proof. Module numbers refer to the cached source files.

Modules	Mathematical role and boundary
02–03	Form bounds, strict-binding compactness, all-sector baseline; exact conditional localization; coherent upper packets and Neumann lower cubes with the same kinetic coefficient. These modules were read in full.
04–05b	Count bootstrap, conditional TF patch, noisy unordered observations, two different expectation towers, and maxima of weak subsolutions. Read in full; their stated comparison estimates are the inputs to our quantitative deductions.
06–09	Scale-free counts in nuclear voids, deterministic nonlinear barriers, inverse conditional comparison, outward propagation, and the terminal charge contradiction. Selected statements and the terminal averaging/charge proof were read; substantial earlier proofs remain imported.
10–11	Atomic distance geometry, simultaneous inverse comparison, common master kernels and outward deficit. Initial statements and interfaces were read; most propagation proofs remain imported.
12–15	Zero-offset deletion supplies a bounded neutral offset; distant insertion gives a positive gap; shell-mass bias gives a C/R tail; ordered asymptotic interfaces. These modules were read in full.

No source error is asserted in these mechanisms. In particular, the source already proves compact signed Coulomb duality and the positive noncompact extension. The new main increment is the optimization and composition of its actual local error budget, not that duality, not an alternative quantum Hamiltonian, and not a proof of the global claims.

1.2 Primary literature and version scope

The assigned preprint has a September 24, 2026 face date. The locally captured text is authenticated against the captured October 8 tree; this is byte provenance, not peer review or live-branch verification. An October 9 primary-source refresh located the publisher's family-263 overview, which advertises the assigned claim and separate generalized energy/radius companions. Those announcements are not independent verification [1].

Hundertmark–Pattakos–Schulz's April 25, 2025 preprint states $N_c(Z) < 1.1185Z + O(Z^{1/3})$ for atoms [2]. Solovej's published Hartree–Fock theorem gives uniform ionization and radius bounds within that approximation, not for every correlated wavefunction in (1) [3]. Lewin's 2025 primary survey records the charged-quantum-particle problem context [4]. Most recently, Chen–Wu's October 7, 2026 v1 abstract claims, in the different two-center Thomas–Fermi–Dirac–von Weizsäcker model, $N \le Z + CZ^{153/154}(1+\log Z)^{3/8}$ uniformly in the two centers and charges [5]. These external statements and metadata were checked; their complete proofs were not audited. No exhaustive priority or present-best-bound claim is needed for the local results below.

2 The actual patch and its energy ledger

Keep the source normalization

$$c_{\mathrm{TF}} = \frac{3}{10}(3\pi^2)^{2/3}, \qquad k = \left(\frac{5}{3}c_{\mathrm{TF}}\right)^{-3/2} = \frac{2^{3/2}}{3\pi^2}, \qquad c_F = 4\pi k.$$

The coarse fermionic kinetic inequality has a different, unspecified constant; substituting it for c_{TF} would invalidate the comparison. In the molecular geometry $a_y = \sup\{s : \nu(B(y, Ls)) \le s^{-3}\}$, with $L = 10^6$. In the atomic geometry $a_y = |y|/L$, $L = 10^5$, $y \neq 0$. The latter has no lower bound at the Thomas–Fermi length $Z^{-1/3}$. Set

$$a = a_y, \quad m_0 = \max(a^{-3}, 1), \quad m = m_0 + \sqrt{\delta a}, \quad e = m^2/a. \tag{3}$$

The source's count bootstrap supplies a universal bound $1 \le P \le C$ on the normalized local second moments, uniformly for every finite δ-state. It is an actual quantum input, not a conclusion of a finite moment fit.

Fix $0 < \beta < 1/100$ with $\beta^{-2}/(am) \le \sqrt{P}$, and set $b = \beta a$. The source constructs a deterministic $t_0 \in [5a, 6a]$ and a smooth square cut, full out on $B(y, t_0)$, supported in $B(y, t_0 + b)$, with gradients $O(b^{-1})$. Its datum X records the out particles' positions and spins. The normalized surviving core is still an antisymmetric wavefunction; write its density ρ_X^c and $\Phi_X = V - K * \rho_X^c$. On $\Omega = B(y, t_0 - 4b)$ let $\rho = \rho_X$ be the unique minimizer of

$$\mathcal{T}_X(q) = c_{\mathrm{TF}} \int_\Omega q^{5/3} - \int_\Omega \Phi_X q + D(q), \qquad q \in L^{5/3}(\Omega), \quad q \ge 0, \tag{4}$$

extended by zero outside Ω. Then $W = \Phi_X - K * \rho$ and $\rho = k W_+^{3/2}$. No total-mass constraint is placed on this patch minimization.

Retain actual out centers with $|x_i - y| < t_0 - 7b$. Their fine density is

$$\sigma = \sum_{i \text{ retained}} \vartheta_b(\cdot - x_i), \tag{5}$$

$$\vartheta_b(x) = \int_{SO(3)} b^{-6} \prod_{j=1}^{3} (b - |(Qx)_j|)_+ \, \mathrm{d}Q.$$

Haar measure has mass one. This is a radial probability kernel supported in $B(0, \sqrt{3}b)$. Thus $\operatorname{supp} \sigma \subset \Omega$; it is not a posterior mean substituted for the retained particles. Define the nonnegative gap

$$\begin{aligned} \mathcal{G}_X(\sigma) &= D(\sigma - \rho) + \mathcal{L}_X(\sigma) + \int_\Omega W_- \sigma, \\ \mathcal{L}_X(\sigma) &= c_{\mathrm{TF}} \int_\Omega \left[\sigma^{5/3} - \rho^{5/3} - \tfrac{5}{3} \rho^{2/3} (\sigma - \rho) \right]. \end{aligned} \tag{6}$$

The last penalty cannot be discarded when later transferring negative fields under exceptional-data laws.

Lemma 2.1 (Exact TF gap, already implicit in the source)**.** *For the preceding actual patch,* $\mathcal{G}_X(\sigma) = \mathcal{T}_X(\sigma) - \mathcal{T}_X(\rho)$.

Proof. Expand the Coulomb quadratic form: $D(\sigma) - D(\rho) = D(\sigma - \rho) + \int (K * \rho)(\sigma - \rho)$. After subtracting the kinetic tangent, the remaining linear term is $[\frac{5}{3} c_{\mathrm{TF}} \rho^{2/3} - W](\sigma - \rho) = W_-(\sigma - \rho)$. Since $W_- \rho = 0$, this is $W_- \sigma$. All pairings are finite for compact $L^{5/3}$ densities and $\Phi_X \in L^{5/2}(\Omega)$. □

The source's comparison lemma, with universal constants, states

$$\mathbb{E} \mathcal{G}_X(\sigma) \le C \left[\delta + (\epsilon + \sqrt{\beta}) Pe + \epsilon^{-3/2} \beta^{1/2} a^{-7} + b^{-1} \sqrt{P} m + b^{-2} (P^{2/3} m^{4/3} + \sqrt{P} m) \right] \tag{7}$$

for every sufficiently small $\epsilon > 0$. If the local nuclear measure on $B(y, 20a)$ is zero, the stronger version is

$$\mathbb{E} \mathcal{G}_X(\sigma) \le C \left[\delta + \sqrt{\beta} Pe + b^{-1} \sqrt{P} m + b^{-2} (P^{2/3} m^{4/3} + \sqrt{P} m) \right]. \tag{8}$$

In both cases $\mathbb{E} \int \sigma^{5/3}$ is bounded by CPe plus the right side, and $T_o \le C(Pe + b^{-2} \sqrt{P} m)$. The cut is full out on $B(y, 4a)$. These facts will also preserve the raw-density estimate.

The origin of each term matters. Shell averaging gives $\mathbb{E} n_{\mathrm{del}}^2 \le C \beta P m^2$. Its interaction with the conditional far field costs $\sqrt{\beta} Pe$. The upper packet's nuclear loss costs $\epsilon^{-3/2} \beta^{1/2} a^{-7}$ by a local $L^{5/2}$ estimate and Young's inequality. The b^{-1} term is Coulomb self-energy; the b^{-2} terms are finite-cube kinetic and localization costs. The matching c_{TF} comes from coherent-state filling with two spins and a Neumann lattice count, not from an arbitrary finite matrix.

3 Optimized local screening and its exact limitation

Theorem 3.1 (Optimized source patch)**.** *Assume the actual source patch construction, count bound and comparison* (7)–(8)*. For sufficiently small universal $a > 0$, the following choices are admissible:*

Patch	*Offset restriction*	β, b	ϵ
General	$\delta \le a^{-7+1/11}$	$a^{5/11}$, $a^{16/11}$	$a^{1/11}$
Nucleus-free	$\delta \le a^{-7+1/5}$	$a^{2/5}$, $a^{7/5}$	*not needed*

With $\kappa = 1/11$ or $1/5$, respectively,

$$\mathbb{E}\mathcal{G}_X(\sigma) \le Ca^{-7+\kappa}, \qquad \mathbb{E}\int \sigma^{5/3} \le Ca^{-7}, \qquad \int_{B(y,4a)} \rho_\psi^{5/3} \le Ca^{-7}. \tag{9}$$

The expectation and the conditional density are those of the newly chosen cut; this is not a bound for an unchanged old packet width.

Proof. For $0 < a < 1$ and $\delta \le a^{-7+\kappa}$, $a^{-3} \le m \le 2a^{-3}$ and $e \le 4a^{-7}$. Write $\beta = a^\theta$, $b = a^{1+\theta}$ and $\epsilon = a^\eta$. On dividing (7) by a^{-7} the non-offset terms are bounded by constants times powers with gains

$$\eta, \quad \theta/2, \quad \theta/2 - 3\eta/2, \quad 3 - \theta, \quad 1 - 2\theta, \quad 2 - 2\theta. \tag{10}$$

At $(\theta, \eta) = (5/11, 1/11)$ these are $1/11, 5/22, 1/11, 28/11, 1/11, 12/11$. Every term is at most a constant times $a^{1/11}$. In (8) only $\theta/2, 3 - \theta, 1 - 2\theta, 2 - 2\theta$ remain; at $\theta = 2/5$ their minimum is $1/5$.

The admissibility condition is not lost in this balance: $\beta^{-2}/(am) \le a^{2-2\theta}$, which tends to zero for both choices, while $\sqrt{P} \ge 1$. Also $\beta < 1/100$ and the required smallness of ϵ hold after decreasing the universal threshold. The density bound follows from the source's bound following (7). Finally $b^{-2}m \le 2a^{-5-2\theta} \le 2a^{-7}$ because $\theta < 1$. Thus $T_o \le Ca^{-7}$, and the coarse kinetic inequality applied to the out density, equal to ρ_ψ on $B(y, 4a)$, proves the last claim. □

Proposition 3.2 (Optimality within this ledger)**.** *Among fixed positive monomial choices $\beta = a^\theta$, $\epsilon = a^\eta$ that control every displayed term separately, the largest general gain is $1/11$. The nucleus-free maximum is $1/5$. Multiplying these choices by fixed positive constants does not change the maximal leading power.*

Proof. A general gain $\kappa > 0$ requires $\eta \ge \kappa$ and $\theta/2 - 3\eta/2 \ge \kappa$, hence $\theta \ge 5\kappa$. The cube term requires $1 - 2\theta \ge \kappa$, so $11\kappa \le 1$. In the source-free case $\theta/2 \ge \kappa$ and $1 - 2\theta \ge \kappa$ give $5\kappa \le 1$. Theorem 3.1 attains both endpoints and checks every other term and geometric constraint. Fixed constants cannot repair a negative power of a as $a \downarrow 0$. □

This is an exact small linear optimization of a source-specific error budget, not optimality of screening itself. Better kinetic remainders, better shell control, or a different upper-packet nuclear estimate may change the feasible region. The source uses $\theta = 1/5$, $\eta = 1/50$ and retains the gain $1/50$; those choices are valid and conservative. Our stronger accuracy also imposes a stricter offset ceiling: for $a < 1$, $a^{-7+1/11} < a^{-7+1/50}$. It is not a free improvement at all of the source's old offsets.

For a concrete exponent comparison let $a = 2^{-110}$, disregarding the unknown universal smallness threshold. The source's normalized discrepancy power is $2^{-11/5}$, whereas the general optimized power is 2^{-10} and the nucleus-free power is 2^{-22}. Constants are not specified, so this illustration is not a numerical error guarantee at that particular scale. The algebraic gain persists asymptotically under the same universal comparison constants.

4 From Coulomb energy to the unsmoothed field

We first specify the physical Hilbert space. The completion of $C_c^\infty(\mathbb{R}^3)$ in $\|\nabla v\|_2$ is $\dot{H}^1(\mathbb{R}^3)$, realized in L^6. For real compactly supported $f \in L^{5/3}$, the source proves

$$u = K * f \in \dot{H}^1(\mathbb{R}^3), \quad -\Delta u = 4\pi f, \quad D(f) = \tfrac{1}{2}\int f(K*f) = \frac{\|\nabla u\|_2^2}{8\pi}, \quad \left|\int fv\right| \le \sqrt{\frac{D(f)}{2\pi}}\|\nabla v\|_2. \tag{11}$$

Compact support supplies $L^{6/5}$ integrability. The source separately extends this to nonnegative $f \in L^{5/3} \cap (\dot{H}^1(\mathbb{R}^3))^*$ by monotone Newton potentials and a positive dual extension. It does not silently identify every noncompact signed distribution with an absolutely convergent Newton integral. Our signed fields below have compact support.

Proposition 4.1 (Explicit pointwise interpolation)**.** *For such f, put $F = \|f\|_{5/3}$, $E_f = D(f)$ and*

$$c_0 = (5\pi^2/4)^{2/5}, \qquad C_0 = \left[(5/2)^{2/7} + (2/5)^{5/7}\right] c_0^{5/7} 2^{1/7}.$$

Then the continuous Newton potential satisfies, at every x,

$$\begin{aligned} |K*f(x)| &\le c_0 F r^{1/5} + \sqrt{2E_f/r} \quad (r>0), \\ \|K*f\|_\infty &\le C_0 F^{5/7} E_f^{1/7}. \end{aligned} \tag{12}$$

If f is random on a probability space, with compact support for each sample and $\mathbb{E}\|f\|_{5/3}^{5/3} \le M_5$, $\mathbb{E}D(f) \le M_D$, then

$$\mathbb{E}\|K*f\|_\infty \le C_0 M_5^{3/7} M_D^{1/7}. \tag{13}$$

No common support radius is needed. Joint measurability is assumed; continuity lets the supremum be taken on a countable dense set.

Proof. For a fixed x use $v_{x,r}(z) = \min(r^{-1}, |x-z|^{-1})$. This belongs to $\dot{H}^1(\mathbb{R}^3)$ with gradient norm squared $4\pi/r$. The remaining singular kernel has

$$\int_{|z|<r} (|z|^{-1} - r^{-1})^{5/2}\,\mathrm{d}z = 4\pi r^{1/2}\int_0^1 s^{-1/2}(1-s)^{5/2}\,\mathrm{d}s = \frac{5\pi^2}{4} r^{1/2}.$$

Hölder and (11) give the first bound. If $F, E_f > 0$, differentiate $Ar^{1/5} + Br^{-1/2}$ with $A = c_0 F$, $B = \sqrt{2E_f}$. Its unique minimum occurs at $r^{7/10} = 5B/(2A)$, giving the stated constant. If either $F = 0$ or $E_f = 0$, then $f = 0$ as a distribution and both conclusions hold. For the expectation, write $F^{5/7}E_f^{1/7} = (F^{5/3})^{3/7} E_f^{1/7} \cdot 1$ and apply Hölder with exponents $7/3, 7, 7/3$. □

This is a classical potential interpolation argument with its constants and source-compatible integrability made explicit. Energy alone cannot bound point evaluation. Indeed, for any nonzero nonnegative smooth compact f, the dilation $f_r(z) = r^{-5/2} f(z/r)$ has $D(f_r) = D(f)$ but $K*f_r(0) = r^{-1/2} K*f(0) \to \infty$. Both factors in (12) are therefore substantive; a singular point probe is not a bounded $\dot{H}^1(\mathbb{R}^3)$ test.

Lemma 4.2 (Density moment recovered from the gap)**.** *For the actual patches in Theorem 3.1,*

$$\mathbb{E}\int \rho^{5/3} + \mathbb{E}\int |\rho - \sigma|^{5/3} \le C a^{-7}. \tag{14}$$

Consequently $\mathbb{E}\|K*(\rho-\sigma)\|_\infty \le C a^{-4+\kappa/7}$.

Proof. For $p = 5/3$, the scalar Bregman remainder is $B_p(s,q) = s^p + (p-1)q^p - pq^{p-1}s$. Young's inequality with a sufficiently small coefficient on q^p gives $B_p(s,q) \geq (p-1)q^p/2 - C_p s^p$ for $s, q \geq 0$. Integrate, use $\mathcal{L} = c_{\mathrm{TF}} \int B_p$ and Theorem 3.1, then $|q-s|^p \leq 2^{p-1}(q^p + s^p)$. In (13) take $M_5 = Ca^{-7}$, $M_D = Ca^{-7+\kappa}$. The exponent is $-3 - 1 + \kappa/7$, as claimed. □

4.1 The two laws are not interchangeable

Let P be the original eigenstate's position-and-spin law, augmented by the source's independent compact noises. At each observation scale the labels are forgotten separately. For the decreasing observation sigma-field $\mathcal{F}_j$, let $\mu_j(z) = \mathbb{E}_P[\sum_i \mathcal{K}_{x_i}(z) \mid \mathcal{F}_j]$, where $\mathcal{K}_x = g_{t_x}^2(\cdot - x)$ is a radial probability packet. Let $A \in \mathcal{F}_j$, $P(A) > 0$, and $Q = P(\cdot \mid A)$, extended by fresh localization labels. The source's event multiplier gives the actual tilted wavefunction

$$\Psi_A = \sqrt{P(A \mid x)/P(A)}\,\Psi, \qquad q[\Psi_A] - E \leq D_0 + C\ell_j^{-2}\log^5(e/P(A)), \quad D_0 = E_n - E.$$

This finite offset must satisfy the chosen patch ceiling; it is not set to zero. No count-dependent extra factor is inserted in this dimension-free likelihood estimate.

Under Q, X is the fresh patch datum, with its own conditional core density ρ_X^c. Decompose $V = V_s + V_c$ as in the source, and set

$$H_j = V_s - K * \mu_j, \qquad h_X = V_s - K * \rho_X^c - K * \rho_X.$$

Write $P^{\mathrm{raw}}, P^{\mathrm{ms}}, P^c, P^{\mathrm{ret}}, P^{\mathrm{del}}$ for raw, master-smeared, core, retained and deleted electron potentials. At a deterministic point where V_s is finite the exact two towers are

$$\mathbb{E}_Q K * \mu_j = \mathbb{E}_Q P^{\mathrm{ms}}, \qquad \mathbb{E}_Q K * \rho_X^c = \mathbb{E}_Q P^c.$$

The first uses $A \in \mathcal{F}_j$ under the original law P; the second conditions the newly localized tilted law on X. Thus

$$\mathbb{E}_Q(H_j - h_X) = \mathbb{E}_Q\left[P^{\mathrm{raw}} - P^{\mathrm{ms}} - P^{\mathrm{del}} - (P^{\mathrm{ret}} - K * \sigma) + K * (\rho_X - \sigma)\right]. \tag{15}$$

This is an identity of expectations, not of individual conditional fields. Finiteness is established in the proof below before subtraction.

Theorem 4.3 (Direct-field transfer with no inverse master-width loss)**.** *Assume the two-law construction above. For its actual tilted patch at y, suppose (9) holds with $\kappa > 0$ and $\beta = a^\theta$; the raw density bound there holds as well. Assume the master widths have a positive fixed-system minimum and Lipschitz constant at most $1/2$, $t_y/a = \tau \leq 1/10$, and $\beta \leq 1/10$. Deleted particles are at distance at least ca from y, and $\mathbb{E}_Q n_{\mathrm{del}} \leq Ca^{-3}\sqrt{\beta}$. Then*

$$|\mathbb{E}_Q(H_j - h_X)(y)| \leq Ca^{-4}\left(\tau^{1/5} + \beta^{1/5} + \sqrt{\beta} + a^{\kappa/7}\right). \tag{16}$$

In particular the general optimized patch gives an $O(a^{1/77})$ normalized error whenever $\tau \leq Ca^{5/77}$, and the nucleus-free optimized patch gives $O(a^{1/35})$ whenever $\tau \leq Ca^{1/7}$. No lower bound on τ is required for this inequality.

Proof. Hölder and the raw density estimate yield, for $0 < u \leq 4a$,

$$\mathbb{E}_Q \sum_{|x_i - y| < u} |x_i - y|^{-1} \leq Ca^{-21/5}u^{1/5} = Ca^{-4}(u/a)^{1/5}.$$

The complementary potential is at most $n/(4a)$ for the fixed finite sector. By Newton's radial comparison, master smearing changes a point potential only when $|x_i - y| \le t_{x_i}$. Lipschitz continuity gives $|x_i - y| \le 2t_y$ in that case. Hence the first difference in (15) has mean at most $Ca^{-4}\tau^{1/5}$. The fine-kernel support is $\sqrt{3}b$, so its corresponding mean loss is at most $Ca^{-4}\beta^{1/5}$. The deleted contribution is bounded by $(ca)^{-1}\mathbb{E}_Q n_{\mathrm{del}}$. Finally Lemma 4.2 bounds the mean absolute value of the last term, indeed its spatial supremum, by $Ca^{-4+\kappa/7}$. The raw, smeared, and core terms are integrable by the preceding near/far split; the retained and comparison potentials are integrable by the same split and the last bound. Thus (15) is legitimate and proves (16).

For the general patch, $\theta/5 = 1/11 > 1/77$ and $\theta/2 = 5/22 > 1/77$. For the nucleus-free patch, $\theta/5 = 2/25 > 1/35$ and $\theta/2 = 1/5 > 1/35$. Substitution of the two upper bounds on τ proves the last claims. □

The source's transfer proof uses the same truncated Newton test but separately bounds its two local density remainders. Its displayed bound contains $a^{\kappa/2}\tau^{-1/2}$ and local TF pointwise hypotheses. Our expected global patch $L^{5/3}$ bound instead optimizes the test radius independently of the master width. The resulting estimate keeps the actual laws and kernels, removes that inverse-width loss, and does not require the separate local nuclear upper bound on ρ_X for the transfer itself. For general hypothetical comparison densities one must supply (14); it is not automatic from energy alone.

This flexibility is local only. A very small master width worsens derivative bounds, spatial nets and observation matching elsewhere. The source's common clamped master kernel, likelihood scale ℓ_j, rare-event thresholds, barriers and sequential choices cannot be replaced merely by choosing $t_y = a^{1+5/77}$ at one point. Under the original master widths, (16) is still valid when its premises hold, but its first term can remain the dominant one.

5 A whole-space certificate for the same TF gap

The original private TR250 proves a bounded-domain Dirichlet hypercircle identity with an equilibrated flux. Its cross term vanishes because of the weak divergence equation and the physical boundary condition. The physical-metric chapter similarly distinguishes a coefficient matrix from its realized energy norm. We now apply those principles on the actual whole space; no artificial zero boundary on a finite ball is imposed. The following is classical convex duality specialized to (4), not a new existence theorem.

Theorem 5.1 (Primal–dual patch enclosure)**.** *Fix a bounded patch Ω, a real $\Phi \in L^{5/2}(\Omega)$, and the minimizer ρ of (4). For any $s \ge 0$ in $L^{5/3}(\Omega)$, extended by zero, and any real $v \in \dot{H}^1(\mathbb{R}^3)$, define*

$$L(v) = -\frac{\|\nabla v\|_2^2}{8\pi} - \frac{2}{5}k\int_\Omega (\Phi - v)_+^{5/2}, \tag{17}$$
$$B_\Phi(s,v) = \int_\Omega \left[c_{\mathrm{TF}} s^{5/3} - (\Phi - v)s + \tfrac{2}{5}k(\Phi - v)_+^{5/2}\right].$$

The integrand of B_Φ is nonnegative, and

$$\begin{aligned} 0 \le \mathcal{T}(s) - \mathcal{T}(\rho) &\le \mathcal{T}(s) - L(v) \\ &= \frac{\|\nabla(K * s - v)\|_2^2}{8\pi} + B_\Phi(s,v). \end{aligned} \tag{18}$$

If $q \in L^2(\mathbb{R}^3;\mathbb{R}^3)$ satisfies $-\operatorname{div} q = 4\pi s$ in whole-space distributions, the first norm in (18) may be replaced by $\|q - \nabla v\|_2$. More generally set $r = -\operatorname{div} q - 4\pi s \in (\dot{H}^1(\mathbb{R}^3))^*$. Then

$$\mathcal{T}(s) - \mathcal{T}(\rho) \le \frac{\left(\|q - \nabla v\|_2 + \|r\|_{(\dot{H}^1(\mathbb{R}^3))^*}\right)^2}{8\pi} + B_\Phi(s,v). \tag{19}$$

Every norm and the residual dual norm concern the whole space.

Proof. For $t \in \mathbb{R}$ direct minimization over $z \geq 0$ gives

$$\inf_{z\geq 0}(c_{\mathrm{TF}} z^{5/3} - tz) = -\tfrac{2}{5}k(t_+)^{5/2}, \qquad z = k(t_+)^{3/2}.$$

Also (11) and completion of the energy square give $D(s) \geq \int vs - \|\nabla v\|_2^2/(8\pi)$. Applying both inequalities to every competitor proves $\mathcal{T}(\rho) \geq L(v)$. The same square completion at s proves the equality in (18). Its integrals are finite: $v \in L^6$ restricts to $L^{5/2}(\Omega)$ and the other factors have conjugate exponents $5/3$ and $5/2$.

Put $u = K * s$. If q is equilibrated, $q - \nabla u$ is distributionally divergence free. It is orthogonal in L^2 to every gradient from $\dot{H}^1(\mathbb{R}^3)$, first by compact testing and then by completion. Therefore

$$\|q - \nabla v\|_2^2 = \|\nabla u - \nabla v\|_2^2 + \|q - \nabla u\|_2^2.$$

For an imperfect flux, let P_∇ be the orthogonal projection onto the closed gradient space. Weak testing shows $\|P_\nabla q - \nabla u\|_2 = \|r\|_{(\dot{H}^1(\mathbb{R}^3))^*}$. The triangle inequality and contraction of P_∇ give $\|\nabla u - \nabla v\|_2 \leq \|q - \nabla v\|_2 + \|r\|_{(\dot{H}^1(\mathbb{R}^3))^*}$. This proves (19) with the stated normalization. □

For $s = \sigma$ and $\Phi = \Phi_X$, Lemma 2.1 identifies the certified quantity with the complete $D+\mathcal{L}+W_-$ patch gap. If v, q are measurably chosen for each actual datum and the right side is integrable, taking expectation gives an upper bound on that same gap. This supplies a possible independent numerical enclosure of a *given* patch calculation. It neither constructs the quantum state nor proves the uniform estimate (7). An approximate potential $\widetilde{\Phi}$ cannot be substituted silently: for any candidate s, its energy changes by at most $\|\Phi - \widetilde{\Phi}\|_{5/2}\|s\|_{5/3}$, and a minimum comparison also needs a bound for the true minimizer's norm. A small residual on finitely many tests is a lower bound on the dual norm, not an upper one.

5.1 A compact exact negative control

Let $r = |x|$ and define on all of $\mathbb{R}^3$

$$u_1 = (1 - r^2)_+^3, \qquad u_2 = r^2(1 - r^2)_+^3, \qquad f_i = -\Delta u_i/(4\pi).$$

These potentials are compactly supported C^2 functions. Their first derivatives vanish at $r = 1$, so their distributional Laplacians have no surface delta. The sources are bounded compact signed functions, hence belong to the source's $L^{5/3}$ class. Since $u_i \in \dot{H}^1(\mathbb{R}^3)$ and solve the full Poisson equation, uniqueness there gives $K * f_i = u_i$. Their net charges are zero, by the vanishing outer radial derivative. Define the actual Gram matrix

$$G_{ij} = \frac{1}{4\pi}\int_{\mathbb{R}^3} \nabla u_i \cdot \nabla u_j = \int_0^1 r^2 u_i'(r)u_j'(r)\,\mathrm{d}r = \begin{pmatrix} 1536/5005 & 512/15015 \\ 512/15015 & 1536/85085 \end{pmatrix}_{ij}. \tag{20}$$

Thus $D(a_1 f_1 + a_2 f_2) = a^T G a/2$, not $|a|^2/2$. The signed source $f_2 - f_1/9$ has zero measurement against u_1 but

$$D(f_2 - f_1/9) = \frac{1}{2}(G_{22} - G_{12}^2/G_{11}) = \frac{16384}{2297295} > 0. \tag{21}$$

The entries follow by differentiating the two polynomials and integrating monomials on $[0, 1]$. This example is a full-space admissible source, not a point-charge surrogate. It disproves the inference

that one vanishing physical residual measurement certifies zero Coulomb energy. It does not exhibit a quantum many-body density or violate screening.

More generally, a compact radial source of total charge Q has $u(r) = Q/r$ outside its support. If a computational ball of radius R contains that support, the omitted energy is exactly $Q^2/(2R)$. For nonspherical sources this monopole calculation alone need not exhaust the tail. A whole-space flux or certified exterior solution must account for the remaining multipoles. Setting the potential to zero on the ball instead changes the physical problem.

6 Smearing error in the source's integrability class

The source proves radial smearing decreases the Coulomb energy of nonnegative densities. The signed-source version and the following explicit remainder are useful when the density is the discrepancy $\sigma - \rho$. They do not assume an unavailable L^2 bound.

Proposition 6.1 (Signed smearing and its two regularity levels)**.** *Let $f \in L^{5/3}(\mathbb{R}^3)$ be real with compact support, and let η be a radial probability measure supported in $B(0,R)$, $R > 0$. Then*

$$0 \le D(f) - D(\eta * f) \le \tfrac{1}{2}(16\pi/7)^{4/5}(2R)^{7/5}\|f\|_{5/3}^2. \tag{22}$$

If additionally $f \in L^2$, and $m_2 = \int |x|^2 \,\mathrm{d}\eta(x)$, then

$$0 \le D(f) - D(\eta * f) \le \frac{2\pi m_2}{3}\|f\|_2^2. \tag{23}$$

For the source kernel ϑ_b, $R = \sqrt{3}b$, $m_2 = b^2/2$, so the last constant is $\pi b^2/3$.

Proof. Convolution averages translations of $\nabla(K * f)$ in its physical L^2 space. Jensen gives the nonnegative energy difference for signed f. Let $\widetilde{\eta}(A) = \eta(-A)$ and $h = K - K * (\eta * \widetilde{\eta})$. Newton's theorem for radial probability measures gives $0 \le h \le K\mathbf{1}_{B(0,2R)}$. Symmetry and convolution of the quadratic form give

$$D(f) - D(\eta * f) = \tfrac{1}{2}\int f(h * f).$$

One may first smooth f and then pass in $L^{5/3}$ on a common compact set and in Coulomb energy. Young's inequality with exponents $5/4, 5/3, 5/2$ and Hölder imply

$$\left|\int f(h * f)\right| \le \|f\|_{5/3}^2\|h\|_{5/4}, \qquad \|h\|_{5/4} \le (16\pi/7)^{4/5}(2R)^{7/5}.$$

This proves (22).

For the second assertion use the unnormalized Fourier convention $\widehat{f}(\xi) = \int e^{-ix\cdot\xi} f(x)\,\mathrm{d}x$. On smooth compact functions, then by energy approximation,

$$D(f) = \frac{1}{4\pi^2}\int \frac{|\widehat{f}(\xi)|^2}{|\xi|^2}\,\mathrm{d}\xi.$$

If X, Y are independent with law η, radial symmetry implies $\mathbb{E}X = 0$, $\mathbb{E}XX^T = (m_2/3)I$. Hence

$$1 - |\widehat{\eta}(\xi)|^2 = \mathbb{E}[1 - \cos(\xi \cdot (X - Y))] \le \tfrac{1}{2}\mathbb{E}(\xi \cdot (X - Y))^2 = \tfrac{m_2}{3}|\xi|^2.$$

Plancherel, $\int |\widehat{f}|^2 = (2\pi)^3\|f\|_2^2$, gives (23). Each unrotated factor in (5) is the law of the difference of two independent uniform coordinates in an interval of length b, whose variance is $b^2/6$. Summing three coordinates gives $m_2 = b^2/2$; rotation does not change it. □

The 7/5 power is consistent with the natural $L^{5/3}$ scaling. Fix a smooth compact nonzero f and scale $f_R(x) = R^{-9/5} f(x/R)$, $\eta_R(x) = R^{-3}\eta_1(x/R)$ for a smooth nontrivial radial probability density. Then $\|f_R\|_{5/3}$ is fixed, while the nonzero energy loss is exactly $R^{7/5}$ times its loss at unit scale. Consequently a uniform higher power of R at this norm level cannot hold for all such sources. The R^2 conclusion requires the extra L^2 premise; Coulomb cusps in TF densities do not supply it automatically. This distinction is relevant to any attempt to replace the source's $L^{5/3}$ comparison by a smoother finite surrogate.

7 Original Ram connections and nonclaims

Three original local proof readings inform this analysis. The volume-21 physical-metric chapter uses an actual reconstruction J, the physical Gram matrix J^*J, and the omitted residual $(I-P)AJc$. Equations (20)–(21) implement that distinction in Coulomb energy. No coefficient condition number is promoted to a whole-space norm estimate. The volume-1 revised RamCS chapter fixes Fourier normalization, convergence domains and inversion in L^2 rather than an unjustified pointwise Fubini calculation. Its unweighted contour is sufficient for (23); an exponential weight would change the energy norm and the Coulomb tail. Its included chapters were not counted as read. The original TR250 hypercircle proof explains exact equilibrium and the dual norm of an imperfect flux. Theorem 5.1 explicitly replaces its bounded Dirichlet space by $\dot{H}^1(\mathbb{R}^3)$ and proves the new domain-specific orthogonality, rather than assuming Poincaré on $\mathbb{R}^3$ [6, 7, 8].

Earlier local reports M529 and M558 were consulted only through a targeted comparison search. They are not original proof evidence and their existing contributions are not credited again. The present main claims concern the specific conditional patch budget and field-transfer law of M528. The classical duality, interpolation and flux identities are supporting tools; no historical novelty is asserted for them.

7.1 Why the global conclusions remain separate

The source's molecular argument uses more than a small local energy: uniform nuclear-void counts, global positive comparison fields with a strict scale gap, inverse-density implications on rare events, open-set gluing after cancellation of nuclear singularities, and preservation of nonlinear reaction under conditional averaging. Its final compact signed charge measure has positive total mass by a far-sphere comparison. None of these global facts follows from a finite Gram matrix or from Theorem 3.1.

The atomic chain has a further logical order. The deficit is first applied at offset zero; deletion then proves $E_{Z-1} \le E + C$ without assuming a neutral offset. Only afterwards is the deficit applied to neutral ground states. Distant insertion uses the singly ionized atom's unit unscreened charge and the complementary loss state to obtain a positive gap. The final shell bound changes law by the actual shell mass, which is measurable in the revealed out data; this removes an otherwise spurious particle-count factor. The source proves a C/R exterior mass bound and handles the finitely many remaining integer charges uniformly over their whole ground-state spaces by compactness. These are retained source arguments, not local consequences newly certified here.

7.2 Advantages, costs and feasible next steps

The optimized rates quantify previously conservative losses while preserving the same kinetic constant, actual cut support and negative field penalty. Direct-field interpolation permits finer master smoothing without an inverse-width penalty in this one estimate. The TF certificate separates

kinetic Fenchel defect from whole-space field mismatch, providing a meaningful target for a future validated patch computation. Signed smearing control requires only the source's actual integrability, with an explicit stronger option under L^2.

The costs are narrower allowed energy offsets, thinner transition layers and larger derivative constants, unidentified universal constants, and the need for actual expected density bounds. A finite solver must also enclose the conditional core field, flux equilibrium, all spatial tails and the distribution of patch data. We measured no computational speedup and produced no numerical ionization constant or atomic radius.

Three plausible extensions have explicit remaining tasks. First, rebuild the common multiscale master kernels and rare-event nets using the new rates, checking every gradient and probability budget before claiming an end-to-end improvement. Second, retain quantitative smearing gains in the coherent upper trial instead of merely dropping them; a useful lower bound on that gain needs additional structure of the actual TF minimizer. Third, implement (19) with certified exterior multipoles and positive integral enclosures under the actual tilted law. None is completed by the present finite checks.

8 Read scope and reproducibility

The owned read ledger records SHA256 before the fresh displays and after them, each literal displayed interval, and its unread complement. It covers all twenty cached public text endpoints and three original private endpoints: 4,349 of 6,585 public lines and 721 original-private lines were freshly read; 2,236 public lines remain unread. Fourteen public files were read completely. The complete foundations, localization, screening, observations, maximum, neutral sector, ionization-gap, outer-radius and interface modules were read; large portions of molecular barriers, void geometry and outward propagation were not. The source claims are mapped above precisely to avoid mistaking this scoped work for an 84-page global proof audit. Public metadata and abstracts are not recorded as external full-proof reads. Private originals remained local.

The owned exact-check script uses rational arithmetic for the exponent ledger, transfer balance, physical radial Gram calculations, Newton reconstruction, hidden-mode energy and finite law identities. It is a supplement to the written proofs, not quantum-state or infinite-family certification. The final execution receipt gives cases and assertions separately, excludes parent and foreign checks, and hashes the executed script. The owned execution passed 126 exact cases and 381 assertions, with zero numerical cases and no parent or foreign cases included. Build, individual-page visual inspection, contribution assessment and source endpoints are frozen in the owned manifest.

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